MATH155: Exponential and Logarithmic Functions

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Learning Objectives

- Evaluate and graph exponential functions
- Convert between exponential and logarithmic form
- Apply properties of logarithms to simplify and solve
- Solve exponential equations using logarithms

Simplify each expression completely. Show all steps and circle your final answer.

Evaluating exponential expressions

1. Evaluate: 2^2.

$$2^2$$

Answer: ____

2. Evaluate: 2^-2 (negative exponent).

$$2^{-2}$$

Answer: ____

3. Evaluate: 4^2.

$$4^2$$

Answer: ____

4. Evaluate: 4^-1 (negative exponent).

$$4^{-1}$$

Answer: ____

5. Evaluate: 4^2.

$$4^2$$

Answer: ____

6. Evaluate: 3^-2 (negative exponent).

$$3^{-2}$$

Answer: ____

7. Evaluate: 2^4.

$$2^4$$

Answer: ____

8. Evaluate: 3^-1 (negative exponent).

$$3^{-1}$$

Answer: ____

9. Evaluate: 3^2.

$$3^2$$

Answer: ____

10. Evaluate: 5^-2 (negative exponent).

$$5^{-2}$$

Answer: ____

11. Evaluate: 2^2.

$$2^2$$

Answer: ____

12. Evaluate: 5^-3 (negative exponent).

$$5^{-3}$$

Answer: ____

Solving simple exponential equations

13. Solve for x: 3^x = 243. (Express the answer as an integer.)

$$\log_3 243$$

Answer: ____

14. Solve for x: 2^x = 8. (Express the answer as an integer.)

$$\log_2 8$$

Answer: ____

15. Solve for x: 2^x = 16. (Express the answer as an integer.)

$$\log_2 16$$

Answer: ____

16. Solve for x: 2^x = 2. (Express the answer as an integer.)

$\log_2 2$

Answer: ____

17. Solve for x: 3^x = 3. (Express the answer as an integer.)

$\log_3 3$

Answer: ____

18. Solve for x: 5^x = 5. (Express the answer as an integer.)

$\log_5 5$

Answer: ____

Evaluating logarithms

19. Evaluate: log_5(625).

$\log_5 625$

Answer: ____

20. Evaluate: log_10(1000).

$\log_{10} 1000$

Answer: ____

21. Evaluate: log_2(4).

$\log_2 4$

Answer: ____

22. Evaluate: log_5(125).

$\log_5 125$

Answer: ____

23. Evaluate: log_10(1000).

$\log_{10} 1000$

Answer: ____

24. Evaluate: log_2(8).

$$\log_2 8$$

Answer: ____

Properties of logarithms — product rule

25. Condense to a single logarithm: log_2(4) + log_2(2).

$$\log_2 4 + \log_2 2$$

Answer: ____

26. Condense to a single logarithm: log_2(8) + log_2(8).

$$\log_2 8 + \log_2 8$$

Answer: ____

27. Condense to a single logarithm: log_10(10) + log_10(100).

$$\log_{10} 10 + \log_{10} 100$$

Answer: ____

28. Condense to a single logarithm: log_3(3) + log_3(27).

$$\log_3 3 + \log_3 27$$

Answer: ____

29. Condense to a single logarithm: log_2(4) + log_2(4).

$$\log_2 4 + \log_2 4$$

Answer: ____

30. Condense to a single logarithm: log_3(27) + log_3(27).

$$\log_3 27 + \log_3 27$$

Answer: ____

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ANSWER KEY & SOLUTIONS

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Topics: Solving simple exponential equations, Evaluating logarithms, Properties of logarithms — product rule, Evaluating exponential expressions. All answers verified by independent computation.

Solutions

Evaluating exponential expressions

1. Evaluate: 2^2.

2^2

→ *Multiply 2 by itself 2 times.*
→ *2^2 = 4.*

Answer: $2^2 = 4$

2. Evaluate: 2^-2 (negative exponent).

2^{-2}

→ *A negative exponent means take the reciprocal: 2^-2 = 1/2^2.*
→ *= 1/4 = 1/4 (as a fraction).*

Answer: $2^{-2} = 1/4$

3. Evaluate: 4^2.

4^2

→ *Multiply 4 by itself 2 times.*
→ *4^2 = 16.*

Answer: $4^2 = 16$

4. Evaluate: 4^-1 (negative exponent).

4^{-1}

→ *A negative exponent means take the reciprocal: 4^-1 = 1/4^1.*
→ *= 1/4 = 1/4 (as a fraction).*

Answer: $4^{-1} = 1/4$

5. Evaluate: 4^2.

4^2

→ *Multiply 4 by itself 2 times.*
→ *4^2 = 16.*

Answer: $4^2 = 16$

6. Evaluate: 3^-2 (negative exponent).

3^{-2}

→ *A negative exponent means take the reciprocal: 3^-2 = 1/3^2.*
→ *= 1/9 = 1/9 (as a fraction).*

Answer: $3^{-2} = 1/9$

7. Evaluate: 2^4.

$$2^4$$

→ *Multiply 2 by itself 4 times.*
→ *2^4 = 16.*

Answer: $2^4 = 16$

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8. Evaluate: 3^-1 (negative exponent).

$$3^{-1}$$

→ *A negative exponent means take the reciprocal: 3^-1 = 1/3^1.*
→ *= 1/3 = 1/3 (as a fraction).*

Answer: $3^{-1} = 1/3$

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9. Evaluate: 3^2.

$$3^2$$

→ *Multiply 3 by itself 2 times.*
→ *3^2 = 9.*

Answer: $3^2 = 9$

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10. Evaluate: 5^-2 (negative exponent).

$$5^{-2}$$

→ *A negative exponent means take the reciprocal: 5^-2 = 1/5^2.*
→ *= 1/25 = 1/25 (as a fraction).*

Answer: $5^{-2} = 1/25$

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11. Evaluate: 2^2.

$$2^2$$

→ *Multiply 2 by itself 2 times.*
→ *2^2 = 4.*

Answer: $2^2 = 4$

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12. Evaluate: 5^-3 (negative exponent).

$$5^{-3}$$

→ *A negative exponent means take the reciprocal: 5^-3 = 1/5^3.*
→ *= 1/125 = 1/125 (as a fraction).*

Answer: $5^{-3} = 1/125$

Solving simple exponential equations

13. Solve for x: 3^x = 243. (Express the answer as an integer.)

$\log_3 243$

→ *Rewrite in logarithmic form: x = log_3(243).*
→ *Ask: 3^? = 243. Since 3^5 = 243, x = 5.*

Answer: $\log_3 243 = 5$

14. Solve for x: 2^x = 8. (Express the answer as an integer.)

$\log_2 8$

→ *Rewrite in logarithmic form: x = log_2(8).*
→ *Ask: 2^? = 8. Since 2^3 = 8, x = 3.*

Answer: $\log_2 8 = 3$

15. Solve for x: 2^x = 16. (Express the answer as an integer.)

$\log_2 16$

→ *Rewrite in logarithmic form: x = log_2(16).*
→ *Ask: 2^? = 16. Since 2^4 = 16, x = 4.*

Answer: $\log_2 16 = 4$

16. Solve for x: 2^x = 2. (Express the answer as an integer.)

$\log_2 2$

→ *Rewrite in logarithmic form: x = log_2(2).*
→ *Ask: 2^? = 2. Since 2^1 = 2, x = 1.*

Answer: $\log_2 2 = 1$

17. Solve for x: 3^x = 3. (Express the answer as an integer.)

$\log_3 3$

→ *Rewrite in logarithmic form: x = log_3(3).*
→ *Ask: 3^? = 3. Since 3^1 = 3, x = 1.*

Answer: $\log_3 3 = 1$

18. Solve for x: 5^x = 5. (Express the answer as an integer.)

$\log_5 5$

→ *Rewrite in logarithmic form: x = log_5(5).*
→ *Ask: 5^? = 5. Since 5^1 = 5, x = 1.*

Answer: $\log_5 5 = 1$

Evaluating logarithms

19. Evaluate: log_5(625).

$$\log_5 625$$

→ Ask: 5 raised to what power equals 625?
→ 5^4 = 625, so log_5(625) = 4.

Answer: $\log_5 625 = 4$

20. Evaluate: log_10(1000).

$$\log_{10} 1000$$

→ Ask: 10 raised to what power equals 1000?
→ 10^3 = 1000, so log_10(1000) = 3.

Answer: $\log_{10} 1000 = 3$

21. Evaluate: log_2(4).

$$\log_2 4$$

→ Ask: 2 raised to what power equals 4?
→ 2^2 = 4, so log_2(4) = 2.

Answer: $\log_2 4 = 2$

22. Evaluate: log_5(125).

$$\log_5 125$$

→ Ask: 5 raised to what power equals 125?
→ 5^3 = 125, so log_5(125) = 3.

Answer: $\log_5 125 = 3$

23. Evaluate: log_10(1000).

$$\log_{10} 1000$$

→ Ask: 10 raised to what power equals 1000?
→ 10^3 = 1000, so log_10(1000) = 3.

Answer: $\log_{10} 1000 = 3$

24. Evaluate: log_2(8).

$$\log_2 8$$

→ Ask: 2 raised to what power equals 8?
→ 2^3 = 8, so log_2(8) = 3.

Answer: $\log_2 8 = 3$

Properties of logarithms — product rule

25. Condense to a single logarithm: log_2(4) + log_2(2).

$$\log_2 4 + \log_2 2$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_2(4) + log_2(2) = log_2(4 * 2) = log_2(8).*
→ *Since 2^3 = 8, the result simplifies to 3.*

Answer: $\log_2 8 = 3$

26. Condense to a single logarithm: log_2(8) + log_2(8).

$$\log_2 8 + \log_2 8$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_2(8) + log_2(8) = log_2(8 * 8) = log_2(64).*
→ *Since 2^6 = 64, the result simplifies to 6.*

Answer: $\log_2 64 = 6$

27. Condense to a single logarithm: log_10(10) + log_10(100).

$$\log_{10} 10 + \log_{10} 100$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_10(10) + log_10(100) = log_10(10 * 100) = log_10(1000).*
→ *Since 10^3 = 1000, the result simplifies to 3.*

Answer: $\log_{10} 1000 = 3$

28. Condense to a single logarithm: log_3(3) + log_3(27).

$$\log_3 3 + \log_3 27$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_3(3) + log_3(27) = log_3(3 * 27) = log_3(81).*
→ *Since 3^4 = 81, the result simplifies to 4.*

Answer: $\log_3 81 = 4$

29. Condense to a single logarithm: log_2(4) + log_2(4).

$$\log_2 4 + \log_2 4$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_2(4) + log_2(4) = log_2(4 * 4) = log_2(16).*
→ *Since 2^4 = 16, the result simplifies to 4.*

Answer: $\log_2 16 = 4$

30. Condense to a single logarithm: log_3(27) + log_3(27).

$$\log_3 27 + \log_3 27$$

→ *Apply the product rule: log_b(A) + log_b(B) = log_b(A*B).*
→ *log_3(27) + log_3(27) = log_3(27 * 27) = log_3(729).*
→ *Since 3^6 = 729, the result simplifies to 6.*

Answer: $\log_3 729 = 6$